Unsupervised Self-Correlated Learning Smoothy Enhanced Locality Preserving Graph Convolution Embedding Clustering for Hyperspectral Images

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***Abstract*— Hyperspectral image (HSI) clustering is an extremely fundamental but challenging task with no labeled samples. Deep clustering methods have attracted increasing attention and have achieved remarkable success in HSI classification. However, most existing clustering methods are ineffective for large-scale HSI, due to their poor robustness, adaptability, and feature presentation. In this article, to address these issues, we introduce unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering (S^2LGCC) for large-scale HSI. Specifically, the spectral-spatial transformation is introduced to transform the original HSI into a graph while preserving the local spectral features and spatial structures. After that, a locality preserving graph convolutional embedding encoder is designed to learn the hidden representation from the graph, in which the deep layer-wise graph convolutional network (LGAT) is proposed to preserve the adaptive layerwise locality features. In addition, the self-correlated learning smoothy module is developed to learn the smoothy information and the nonlocal relationship in the hidden representation space for clustering. Finally, a self-training strategy is proposed to cluster the graph node, in which a self-training clustering objective employs soft labels to supervise the clustering process. The proposed S^2LGCC is jointly optimized by the fusion graph reconstruction loss and self-training clustering loss, and the two benefit each other. On Indian Pines (IP), Salinas, and UH2013 datasets, the overall accuracies (OAs) of our S^2LGCC are 71.76%, 82.61%, and 63.82%, respectively.**

***Index Terms*— Hyperspectral image (HIS) clustering, layer-wise graph convolutional network (LGAT), self-correlated learning smoothy, self-training.**

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I. Introduction

HYPERSPECTRAL images (HSIs) captured by hyperspectral sensors contain hundreds of continuous spectral bands, with which the detailed spatial-spectral information of landcover can be provided [1], [2], [3]. Due to its special characteristics, his has attracted increasing attention and has been widely applied in military reconnaissance, geological mapping, biochemical detection, forest fire detection, and target recognition [4], [5], [6], [7].

HSI classification (HSIC), aiming to classify the pixels contained in HSIs to predefined category labels for the corresponding landcover, is the fundamental technique for the aforementioned applications [8]. In the past decades, many HSIC algorithms, e.g., multiple feature extraction [9], kernel classifier learning [10], sparse representation [11], [12], spatial-spectral integration [13], and deep learning techniques [14], have been designed to explore the underlying features of HSIs and improve the HSIC accuracy. Although the above algorithms have optimized and improved classification results at different levels, they are all supervised or semisupervised methods relying on annotated samples to classify HSIs [15]. In practice, it is a high labor-intensive and time-consuming progress to labhisHSI pixels, which is a crucial limitation that must be considered in the application of the HSIC methods [16]. To solve the aforementioned problem caused by labeled samples deficiency, the unsupervised clustering method, aiming to group similar pixels into different clusters with no prior labels, has attracted increasing attention. Compared with supervised or semisupervised methiss, HSI clustering is often a fundamental but challenging task, due to prior knowledge deficiency, large spectral variability, and high-dimension of HSIs [17].

In the early stage, many traditional clustering methods have been adohisd for HSI clustering, including k-means [18], independent component analysis (ICA) model [19], artificial immune clustering (AIC) [20], spectral clustering (SC) [21], and fuzzy C-means (FCM) [22]. These methods were originally designed for natural images; thus, their clustering performances for HSIs with complex characteristics are limited. Recently, we can mainly divide the existing traditional clustering methods into two categories, i.e., spectral-only and spectral-spatial methods. The spectral-only methods extract the latent feature representations of HSIs in the spectral

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domain to cluster the pixels. For example, in [23], an artificial immune network was phisosed for HSI clustering by simulating the inherenhistructure of HSI. Paoli *et al.* [24] designed a clustering method named multiobjective partical swarm optimization (MOPSO) aiming at solving the problems of feature extraction, class statistical parameters estimation, and class number estimation simultaneously. These methods have promoted the application of unsupervised clustering in HSIC. However, spectral-only methods simply extract the spectral information of HSIs, while the spatial information contained in neighboring pixels is ignored, which limited their clustering performances. To overcome the shortcomings of spectral-only methods and obtain more discriminahise features for HSI clustering, spectral-spatial clustering methods were proposed, in which spectral information and spatial information were jointly utilized. For instance, Chen and Zhang [25] proposed an improved FCM clushising algorithm for HSI by a using kernel-induced distance measure. To preserve the global and local structures of HSIs, in [26], a dual-space transfer learning algorithm is proposed. In addition, to solve the high-dimensional problem, some sparse subspace clustering (SSC) algorithms [27], [28], [29] have appeared, which can cluster the pixels in the low-dimensional subspace by learning the coefficient representation matrix and constructing the affinity matrix. Zhang *et al.* [28] proposed a novel spectral–spatial sparse subspace clusterhis (S4C) algorithm for HSI, in which the high spectral correlation and rich spatial information of the HSIs are taken into consideration. Huang *et al.* [27] developed a novel graph convolutional sparse subspace coclustering (GCSSC) model with nonnegative orthogonal factorization for large HSIs, in which affinity matrix learning and spectral coclustering are integrated into a unified optimizing model to obtain the optimal clustering results.

The aforementioned methods are shallow clustering methods based on the hand-crafted feature or raw feature spaces and cannot learn his high-level features of HSI. Therefore, they are not able to present the large spectral variability and the complex internal structure of HSIs effectively [30], [31]. Recently, with the rapid development and wide application of deep learning, some deep learning clustering frameworks have been designed for clustering tasks [32]. Xie *et al.* [31] pioneer introduced the deeply embedded clustering (DEC) method to learn high-level feature representations by adopting deep networks. Then, deep adaptive image clustering (DAC) [33] was proposed, in which the clustering task was transformed into a binary pairwise classification problem. After that, a generative adversarial network (GAN) [34] was emphised to cluster and generate HSI images in the latent space. Inspired by the subspace clustering network, a deep subspace clustering network named DSC-Net was proposed by Li *et al.* [35]. As the autoencoder technique successful application in the computer vision field, some autoencoder-based clustering methods have been designed for spatial-spectral feature learning. For instance, Hou *et al.* [36] intrhisced contrastive learning into HSI features learning, and the unlabeled samples were utilized to alleviate the label information insufficient problem. In [37], a spatial-spectral subspace clustering network was designed to extract spatial-spectral features by adopting collaborative constraints and the multiscale autoencoder. Due to the nonlinear mappings thhiscan be learned from the original HSI into the latent representation, deep clustering methods can obtain more robust clustering performances compared with the shallow clustering methods. Nevertheless, most of the deep feature extraction models cannot effectively learhisemarkable and adaptive features for HSI clustering, especially the smooth features in HSIs are ignored; thus, the clustering results improvement are limited. In addition, most of the deep clustering methods excessively extract global features and pay less attention to local features of HSIs, which leads to the deficiency of diversity features for clustering. Finally, most of the existing clustering methods lack an effective solution for network training without labeled samples [38], [39], [40].

To overcome these drawbacks, we design a novel unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering (S^2LGCC) for HSIs. Specifically, we first introduce a spechisl-spatial transformation mechanism for HSI preprocessing to transform HSIs into graph while preserving the local spectral features and spatial structure. After that, to learn a more representative hidden representation, a locality preserving graph convolutional embedding encoder is proposed, in which the deep layer-wise graph convolutional network (LGAT) is proposed to learn the locality features by minimizing the hybrid graph reconstruction loss. In addition, to learn the smoother features in hidden representation for subsequent clustering, we design a self-correlated learning smoothy module, with which the smoothy information and the nonlocal relationship in the hidden representation space can be extracted. Finally, a self-training strategy is proposed, in which a self-training clustering objective employs soft labels to supervise the clustering process and obtain appropriate hidden representations for node clustering. In this study, the four aforementioned modules are jointly integrated into a unified end-to-end network by adopting jointly optimizing loss for HSI clustering, and each component benefits the others.

In summary, the main contributions of this study are summarized as follows.

1) We develop a locality preserving graph convolutional embedding autoencoder, with which the layerwise graph features can be preserved, and the remarkable and adaptive features for HSI clustering can be effectively learned by minimizing the hybrid graph reconstruction loss.
2) We design a self-correlated learning smoothy module to extract the smoothy information and nonlocal relationship contained in the hidden representation space for subsequent graph reconstruction and node clustering.
3) We design a self-training clustering mechanism to refine the clustering results, in which soft labels generated from graph embedding are proposed to supervise the self-training clustering process, and the hidden representation is updated by minimizing the Kullback–Leibler (KL) divergence between the target distribution and the soft label's distribution.

The remainder of this article is organized as follows. In Section II, we discuss the related concepts and definitions. Section III introduces the proposed S^2LGCC for HSI

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TABLE I
MAIN NOTIONS AND DESCRIPTIONS

Notations	Meaning
N	Number of nodes contained in the graph.
d, d_i	The dimension of $\boldsymbol{X}$ and the ith latent feature.
c	Number of clusters.
$\boldsymbol{W}$	Learnable weight matrix.
$\boldsymbol{S}_i$	ith superpixel.
$\boldsymbol{X} = \{\boldsymbol{x_1}; \cdots; \boldsymbol{x_N}\}$	Node feature matrix.
$\boldsymbol{E}$	Identity matrix
$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$	Graph $\mathcal{G}$ with edges set $\mathcal{E}$ and nodes set $\mathcal{V}$.
$\boldsymbol{A}, \widehat{\boldsymbol{A}}$	Adjacency matrix of $\mathcal{G}$ and adjacency matrix of a reconstructed graph.
$\boldsymbol{D} = diag(d_1, \cdots, d_n)$	Degree matrix of $\boldsymbol{A}$.
$\boldsymbol{L}$, $\boldsymbol{L}_{sym}$	Graph Laplacian Symmetric and normalized Laplacian of $\mathcal{G}$.
$\boldsymbol{U} = [\boldsymbol{u_0}, \boldsymbol{u_1}, \cdots, \boldsymbol{u_{n-1}}]$	Eigenvector matrix of $\boldsymbol{L}_{sym}$.
σ	Nonlinear activation function.
λ_{max}, λ_i	Largest eigenvalue, ith eigenvalue.
$\boldsymbol{\Lambda}, \boldsymbol{\Lambda_{ii}} = \lambda_i$	Eigenvalues diagonal matrix.
$\cdot\|\|\cdot$	Concatenation operation.
$\boldsymbol{Z_i}$	ith layer encoder output.
$N(\cdot)$	Neighbourhood of $\bullet$.
$\odot$	Element-wise product.
P, Q	The target distribution and soft labels distribution.
η	Learning rate.

clustering. Section IV reports the experimental results on the effectiveness of the proposed S^{2}LGCC. In Section V, we present the conclusions, and the study limitations and future work directions are discussed.

II. RELATED CONCEPTS AND DEFINITIONS

A. Notations and Descriptions

Given an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$ with c different classes (i.e., clusters), where $\mathcal{V} = \{v_i\}_{i=1,\ldots,N}$ and $\mathcal{E} = \{e_{ij}\}_{i,j=1,\ldots,N}$ denote the node set with n nodes and the edge set between node, respectively. The graph $\mathcal{G}$ is characterized by the adjacency matrix $\boldsymbol{A} = (a_{ij})_{N\times N} \in \mathbb{R}^{N\times N}$ and attribute matrix $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$. Here, a_{ij} is the edge weight between node i and node j, $a_{ij} = 1$, if $\{v_i, v_j\} \in \mathcal{E}$, otherwise $a_{ij} = 0$. $\boldsymbol{x}_i \in \mathbb{R}^d$ is the node feature value associated with the node v_i with d attribute dimension. Furthermore, $\cdot\|\cdot$ denotes the concatenation operation. The main notations adopted in this article and their descriptions are demonstrated in Table I.

B. Convolution on Graph

The graph convolution operation can be explained as both spatial operators [41] and spectral operators [42]. In this article, the convolution operator is defined from the frequency, which presents the graph signals in the spectral domain. Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$, the topological structure of the graph $\mathcal{G}$ can be denoted by constructing an adjacency matrix $\boldsymbol{A}$. The mathematical representation of the normalized graph can be defined as $\boldsymbol{L_{sym}} = \boldsymbol{I_n} - \boldsymbol{D}^{-(1/2)}\boldsymbol{A}\boldsymbol{D}^{-(1/2)}$, where $\boldsymbol{D_{ii}} = \sum_{v_j\in\mathcal{E}} (a_{i,j})$. $\boldsymbol{L}$ is real symmetric positive semidefinite and can be factored $\boldsymbol{L} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U^T}$, where $\boldsymbol{U}\boldsymbol{U^T} = \boldsymbol{E}$. Given a graph signal $\boldsymbol{X} \in \mathbb{R}^N$, the Fourier transform of $\boldsymbol{X}$ can be expressed as $\boldsymbol{\mathcal{F}}(\boldsymbol{X}) = \boldsymbol{U^T}\boldsymbol{X}$, and the inverse graph Fourier transform can be expressed as $\boldsymbol{\mathcal{F}^{-1}}(\widehat{\boldsymbol{X}}) = \boldsymbol{U}\widehat{\boldsymbol{X}}$, where $\widehat{\boldsymbol{X}}$ is the result of Fourier transform. The graph Fourier transform projects the graph feature vector into the orthogonal space, in which the basis is composed of the eigenvector of the normalized graph Laplace operator. The elements in $\widehat{\boldsymbol{X}}$ are the coordinates projected by $\boldsymbol{X}$ in the orthogonal space, and $\boldsymbol{X}$can be expressed as $\boldsymbol{X} = \sum_{\boldsymbol{i}} \widehat{\boldsymbol{x}}_{\boldsymbol{i}}\boldsymbol{u_i}$. We can express the graph convolution of $\boldsymbol{X}$ with a graph filter $\boldsymbol{g}$ as

$$\begin{aligned}\mathbf{g} * \boldsymbol{X} &= \boldsymbol{\mathcal{F}^{-1}}(\mathcal{F}(\boldsymbol{x}) \odot \mathcal{F}(\mathbf{g})) \\ &= \boldsymbol{U}\left(\boldsymbol{U}^T\boldsymbol{X} \odot \boldsymbol{U}^T\mathbf{g}\right).\end{aligned} \tag{1}$$

If the graph filter is expressed as $\boldsymbol{g}_\theta = \mathrm{diag}(\boldsymbol{U}^T\mathbf{g})$, then

$$\boldsymbol{g}_\theta * \boldsymbol{X} = \boldsymbol{U}\boldsymbol{g}_\theta(\boldsymbol{\Lambda})\boldsymbol{U}^T\boldsymbol{X}. \tag{2}$$

We can define a spectral-based graph convolution algorithms following the definition of (2). The aggregation operation on graph node is calculated as

$$\boldsymbol{X}_{i+1} = \sigma\left(\widetilde{\boldsymbol{D}}^{\frac{1}{2}}\widetilde{\boldsymbol{A}}\widetilde{\boldsymbol{D}}^{-\frac{1}{2}}\boldsymbol{X}_i\boldsymbol{W}_i\right) \tag{3}$$

where $\boldsymbol{X}_0 = \boldsymbol{X}$. $\widetilde{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}$ and $\widetilde{\boldsymbol{D}}_{ii} = \sum_j \widetilde{\boldsymbol{A}}_{i,j}$ are defined as the renormalization terms of $\boldsymbol{A}$ and $\boldsymbol{D}$, respectively, to enhance stability in the process of network training. Equation (3) demonstrates the node feature transformation of GCN from $\boldsymbol{X}_i \in \mathbb{R}^{N\times d_i}$ to $\boldsymbol{X}_{i+1} \in \mathbb{R}^{N\times d_{i+1}}$. $\boldsymbol{W}_i \in \mathbb{R}^{d_i\times d_{i+1}}$ is a learnable weight matrix to realize the linear transformation on the feature matrix.

C. Graph Attention Convolution

Given a graph $\mathcal{G}$, the linear transformation from $\boldsymbol{x}_i \in \mathbb{R}^d$ to $\boldsymbol{x}_i' \in \mathbb{R}^{d'}$ process can be expressed as

$$\boldsymbol{x}_i' = \boldsymbol{W}^T\boldsymbol{x}_i, \quad \boldsymbol{W} \in \mathbb{R}^{d\times d'}. \tag{4}$$

If node v_j is a neighbor of node i, the attention score e_{ij} can be calculated to evaluate the importance of node v_i to node v_j, namely

$$e_{ij} = \left(\sigma\left(\boldsymbol{a}^T\left[\boldsymbol{W}^T\boldsymbol{x_i}||\boldsymbol{W}^T\boldsymbol{x_j}\right]\right)\right) \tag{5}$$

where $||$ denotes the concatenation operation, $\boldsymbol{a}^T \in \mathbb{R}^{2d'}$ is the parameter vector, and σ is the LeakyReLU. Then, we can convert e_{ij} to the following form through a softmax function:

$$\begin{aligned}a_{ij} &= \mathrm{softmax}\left(e_{ij}\right) \\ &= \frac{\exp\left(\sigma\left(\boldsymbol{a}^T\left[\boldsymbol{W}^T\boldsymbol{x}_i||\boldsymbol{W}^T\boldsymbol{x_j}\right]\right)\right)}{\sum_{k\in N_i}\exp\left(\sigma\left(\boldsymbol{a}^T\left[\boldsymbol{W}^T\boldsymbol{x}_i||\boldsymbol{W}^T\boldsymbol{x}_k\right]\right)\right)}.\end{aligned} \tag{6}$$

The final mathematical expression of graph convolution output of GAT is

$$\boldsymbol{x}_i' = \sigma\left(\sum_{j\in N_i} a_{ij}\cdot\boldsymbol{W}^T\boldsymbol{x}_j\right) \tag{7}$$

where a_{ij} is the learned attention weight as (6), and N_i denotes the index of neighbor nodes of node i.

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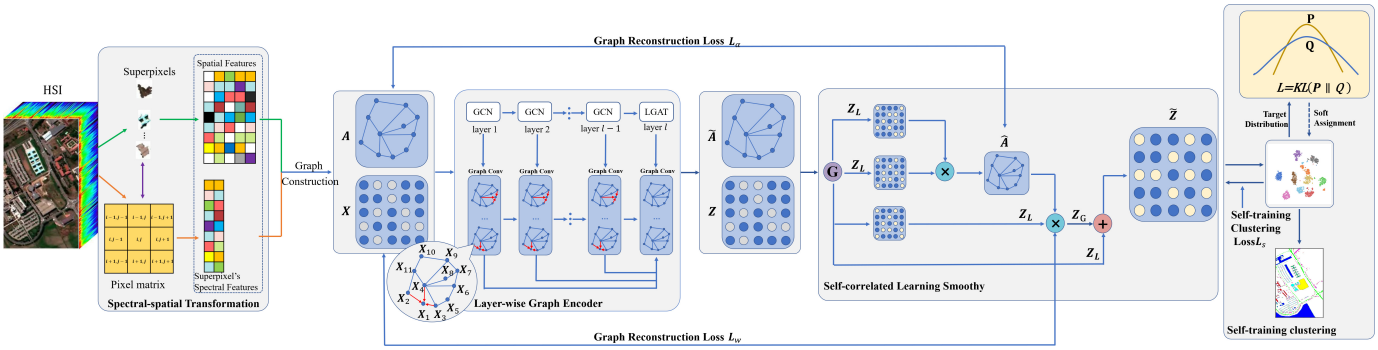

Fig. 1. Illustration of unsupervised self-correlated learning S^{2}LGCC for HSI framework. First, we introduce a spectral-spatial transformation module to transform the original HSI from pixel to locality features preserving graph node; then, the graph is constructed. Subsequently, we develop a locality preserving graph convolutional embedding encoder to learn the hidden representation $\boldsymbol{Z}$. Furthermore, a self-correlated learning smoothy mechanism is designed to extract the smoother features for clustering, and the graph is constructed by adopting $\boldsymbol{L}_\alpha$ and $\boldsymbol{L}_w$. Finally, the self-training clustering is performed. In the flowchart, G denotes message passing, × is matmul product, and + means addition.

III. Proposed Method

In this section, the flow diagram and each step in our method are described in detail. Specifically, the problem definition and overall framework of the proposed method are first introduced to illustrate the problems solved in this article. Then, HSI preprocessing of S^{2}LGCC and the original graph construction mechanism are introduced. Subsequently, we describe the details of unsupervised self-correlated learning S^{2}LGCC, including locality preserving graph convolutional embedding encoder, self-correlated learning smoothy, graph convolutional embedding decoder, joint loss of the proposed network, and training process. Finally, the computational complexity of our network is analyzed.

A. Problem Definition and Overall Framework

In this article, we aim to classify the pixels contained in an HSI into different categories according to their spectral-spatial features. First, we should transform the HSI into a nondirected graph $\mathcal{G}$; then, the node of graph $\mathcal{G}$ will be mapped into low-dimensional embeddings and clustered into c clusters, i.e., $\{C_1, C_2, \ldots, C_c\}$, by using an unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding autoencoder to learn the latent presentation $\boldsymbol{Z}$. Therefore, the nodes in a single cluster contain similar spectral feature values and spatial structure while the farthest distant between different clusters.

The overall framework illustration of S^{2}LGCC is shown in Fig. 1 and consists of four components: spectral-spatial transformation, locality preserving graph convolution embedding encoder, self-correlated learning smoothy, and graph convolution embedding decoder.

1) *Spectral-Spatial Transformation:* The original HSI is transformed into a graph while preserving the local spectral features and spatial structure.
2) *Locality Preserving Graph Convolutional Embedding Encoder:* The node feature and the graph structure of the graph transformed by spectral-spatial transformation are taken as the input. To learn more representative hidden representation $\boldsymbol{Z}$, this module builds a deep LGAT to learn the locality features by minimizing the hybrid graph reconstruction loss $\boldsymbol{L}_c$.
3) *Self-Correlated Learning Smoothy:* The smoothy information and the nonlocal relationship in the hidden representation $\boldsymbol{Z}$ space can be extracted for subsequent graph reconstruction and node clustering.
4) *Graph Convolutional Embedding Decoder:* The proposed method is composed of two decoders. Namely, the graph is reconstructed by minimizing the weighted feature matrix reconstructed loss $\boldsymbol{L}_w$ and the adjacency matrix reconstructed loss $\boldsymbol{L}_\alpha$ contained in the hybrid graph reconstruction loss $\boldsymbol{L}_c$. The self-training clustering can be performed by minimizing the KL divergence between the target distribution $\boldsymbol{P}$ and the soft label's distribution $\boldsymbol{Q}$.

In our S^{2}LGCC, the spectral-spatial transformation, locality preserving graph convolution embedding encoder, self-correlated learning smoothy, and graph convolution embedding decoder are jointed in a unified network for HSI clustering, and each component benefits the others.

B. HSI Preprocessing and Graph Construction

An HSI usually contains rich spectral-spatial information and a large number of pixels, which not only provides sufficient features for clustering but also brings a huge amount of calculation. In practice, the amount of calculation complexity of graph convolutional neural network (GNN) increases exponentially with the increase of the number of nodes. Thus, it is unrealistic to calculate each pixel as a node of the graph. To solve the problem, we first utilize unsupervised PCA [43] to reduce the dimensionality of given HSI, i.e., $I_B = \{x_1, x_2, \ldots, x_N\}$ with N pixels and B channels, and the first principal component is preserved to generate a reduced image $I_b = \{x_1^r, x_2^r, \ldots, x_N^r\}$. After that, a segmentation algorithm named SLIC [44] is adopted to segment the pixels into superpixels, with which the number of graph nodes is greatly reduced while preserving the local spatial features. The mathematical expression of HSI with compact superpixels is

$$\text{HSI} = \cup_{i=1}^{K} S_i, S_i \cap S_j = \emptyset, \quad i \neq j; i, \ j = 1, 2, \ldots, K \quad (8)$$

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where $S_i = \{p_{i,1}, \ldots, p_{i,n_i}\}$ denotes a superpixel containing n_i connected pixels, and K is the total number of superpixels.

As described above, the entire HSI can be segmented into superpixels by the spatial transformer. However, the spectral features are destroyed during the pixel to superpixel transformation process. To recover the spectral features for superpixels, we extract the spectral value of each pixel from the original HSI directly; then, the mean spectral value of pixels in the superpixel is treated as the superpixel's spectral values. The spectral feature vector of the pixel at the spatial location $P_0 = (x, y)$ is

$$\boldsymbol{x}(p_0) = (X_1(P_0), X_2(P_0), \ldots, X_B(P_0)) \tag{9}$$

where $X_i(P_0)$ is the spectral value of the pixel in the ith spectral channel at the spatial location P_0. With (9), the mathematical expression of the HSI feature $\boldsymbol{X}$ with superpixels is

$$\begin{aligned} \boldsymbol{X} &= [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_K]^T \\ &= \left[\frac{1}{n_1}\sum_{k=1}^{n_1}\boldsymbol{x}_k^1, \frac{1}{n_2}\sum_{k=1}^{n_2}\boldsymbol{x}_k^2, \ldots, \frac{1}{n_K}\sum_{k=1}^{n_K}\boldsymbol{x}_k^K\right]^T \end{aligned} \tag{10}$$

where $\boldsymbol{X}_i$ is the feature vector of the ith superpixel.

A graph is defined by the adjacency matrix $\boldsymbol{A}$ and feature matrix $\boldsymbol{X}$. In our proposed method, the superpixels are taken as graph nodes, and their feature matrix $\boldsymbol{X}$ is defined in (10). We define the adjacency matrix $\boldsymbol{A}$ using a radial basis function (RBF) as

$$\boldsymbol{A}_{ij} = \begin{cases} e^{-\rho\|\boldsymbol{X}_i - \boldsymbol{X}_j\|^2}, & \text{if } \boldsymbol{X}_i \in N_t(\boldsymbol{X}_j) \text{ or } \boldsymbol{X}_j \in N_t(\boldsymbol{X}_i) \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where $\rho = 0.2$ is an empirical set to control the RBF width, and $N_t(\boldsymbol{X}_j)$ is the t-hop neighbors set of node $\boldsymbol{X}_j$.

C. Locality Preserving Graph Convolutional Embedding Encoder

The goal of HSI clustering is to cluster the pixels in HSI into belonging clusters with high accuracy. Therefore, how effectively presenting the spectral and structural features in the transformed graph is crucial to our S^{2}LGCC. To represent both node feature $\boldsymbol{X}$ and graph structure $\boldsymbol{A}$ in a unified framework, a variant of the graph convolution network is developed to learn each node's hidden representations by aggregating its neighbors. Given a transformed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$, referring to the propagation rules of GCN in (3), the propagation function of a graph convolutional network layer is formed as

$$\boldsymbol{Z}_{l+1} = \sigma\left(\widetilde{\boldsymbol{D}}^{\frac{1}{2}}\widetilde{\boldsymbol{A}}\widetilde{\boldsymbol{D}}^{-\frac{1}{2}}\boldsymbol{Z}_l\boldsymbol{W}_l\right) \tag{12}$$

where $\widetilde{\boldsymbol{D}}$ is the degree matrix of $\widetilde{\boldsymbol{A}}$, $\boldsymbol{Z}_l = \{\boldsymbol{x}_1^l, \boldsymbol{x}_2^l, \ldots, \boldsymbol{x}_N^l\} \in \mathbb{R}^{N\times d_l}$ is the lth layer learned representation, $\boldsymbol{Z}_0 = \boldsymbol{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$, and $\boldsymbol{W}_l$ is the coefficient learned during the network training. From (12), we can observe that extracting the latent spectral-spatial features from the graph.

As described above, we can adopt a multilayer GCN network to learn each node's spectral-spatial representations by aggregating its neighbors. However, multilayer GCN cannot learn layerwise intermediate information for node clustering. In addition, the available deep clustering methods are usually limited to refine globalization information and neglect the multiscale spatial-spectral features embedded in different layers [31]. The features at different layers contain different levels of semantic description to depict the input data, and the final clustering task may require different locality layerwise information. Accordingly, we should determine the importance of different levels of local features to clustering. To this end, inspired by the graph attention network (GAT), we develop LGAT mechanism to automatically combine different locality layerwise information features by adopting a layerwise mechanism. The conceptual visual interpretation is shown in Fig. 2.

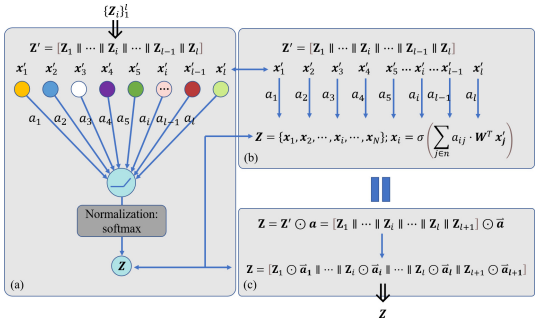

Fig. 2. Conceptual visual interpretation of LGAT. (a) Visual map of LGAT. (b) LGAT is illustrated on a nodewise perspective. (c) LGAT is expressed in a layerwise perspective. Here, the "softmax" is used for normalization. "⇓" is adopted to present the input and output of LGAT.

To preserve the multiscale locality layerwise information contained in each layer, a concatenation manner is preliminarily operated, namely

$$\boldsymbol{Z}' = \left[\boldsymbol{Z}_1 \parallel \cdots \parallel \boldsymbol{Z}_i \parallel \cdots \parallel \boldsymbol{Z}_l \parallel \boldsymbol{Z}_{l+1}\right] \tag{13}$$

where $\boldsymbol{Z}' = \{\boldsymbol{x}_1', \boldsymbol{x}_2', \ldots, \boldsymbol{x}_n'\} \in \mathbb{R}^{N\times d_z}$, where $z = \sum_{i=1}^{l} d_i$, $\boldsymbol{x}_2'$ is the node feature vector aftering concatenation operation.

After that, the LGAT mechanism is introduced to extract the concatenated features $\boldsymbol{Z}'$, where different convolution layer features will be weighted different attention coefficient $\vec{\boldsymbol{a}}$. The mechanism is followed as

$$\begin{aligned} \boldsymbol{Z} = \boldsymbol{Z}' \odot \vec{\boldsymbol{a}} &= \left[\boldsymbol{Z}_1 \parallel \cdots \parallel \boldsymbol{Z}_i \parallel \cdots \parallel \boldsymbol{Z}_l \parallel \boldsymbol{Z}_{l+1}\right] \odot \vec{\boldsymbol{a}} \\ &= \left[\boldsymbol{Z}_1 \odot \vec{\boldsymbol{a}}_1 \parallel \cdots \parallel \boldsymbol{Z}_i \odot \vec{\boldsymbol{a}}_2 \parallel \cdots \parallel \boldsymbol{Z}_l \odot \vec{\boldsymbol{a}}_l \parallel \boldsymbol{Z}_{l+1} \odot \vec{\boldsymbol{a}}_{l+1}\right]. \end{aligned} \tag{14}$$

Referring to (6), the attention coefficient a_{ij} between node i and node j can be formulated as

$$\begin{aligned} a_{ij} &= \operatorname{softmax}(e_{ij}) \\ &= \frac{\exp\left(\sigma\left(\boldsymbol{a}^T\left[\boldsymbol{W}^T\boldsymbol{x}_i' || \boldsymbol{W}^T\boldsymbol{x}_j'\right]\right)\right)}{\sum_{k\in n}\exp\left(\sigma\left(\boldsymbol{a}^T\left[\boldsymbol{W}^T\boldsymbol{x}_i' || \boldsymbol{W}^T\boldsymbol{x}_k'\right]\right)\right)}. \end{aligned} \tag{15}$$

From the spectral values perspective of each node, we can represent (14) as

$$\boldsymbol{x}_i = \sigma\left(\sum_{j\in n} a_{ij}\cdot\boldsymbol{W}^T\boldsymbol{x}_j'\right). \tag{16}$$

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From (16), we note that $\boldsymbol{x}_i$ contains rich layerwise information for different clustering tasks. As described above, (14) and (16) are consistent in principle.

D. Self-Correlated Learning Smoothy

A graph signal (node feature) is smooth if the presentation of neighbor nodes on the graph contains similar features, which contributes to node clustering. As described in Section III-B, we can effectively present the spectral features and structural features of the transformed graph by locality preserving the low-pass graph convolutional embedding encoder. However, the hidden representation $\boldsymbol{Z}$ is not smooth, which is not suit for subsequent clustering. To some degree, the hidden representation $\boldsymbol{Z}$ ignores the local structure within the data. To learn the smooth hidden representation of the graph, we develop the self-correlated learning smoothy module. First, the hidden representation $\boldsymbol{Z}$ is processed by a message passing operation (a graph convolution-like operation), with which the local structure within the graph is enhanced, namely

$$\boldsymbol{Z}_L = \widetilde{\boldsymbol{A}}\boldsymbol{Z}. \tag{17}$$

In (17), we can observe that $\boldsymbol{Z}$ is enhanced by the local structure, namely, $\boldsymbol{Z}_L$ contains the rich spectral and local structure of the graph.

Then, a self-correlated learning mechanism is further introduced to exploit the nonlocal relationship contained in the hidden representation space among nodes. Specifically, a normalized self-correlation matrix $\boldsymbol{S} \in \mathbb{R}^{N\times N}$ is first calculated, namely

$$\boldsymbol{S}_{ij} = \frac{e^{\left(\mathbf{Z}_L\mathbf{Z}_L^T\right)_{ij}}}{\sum_{k=1}^{N} e^{\left(\mathbf{Z}_L\mathbf{Z}_L^T\right)_{ik}}}. \tag{18}$$

With $\boldsymbol{S}$ and $\boldsymbol{Z}_L$ being recombined by taking the global correlation among nodes, the mathematical expression can be expressed as

$$\boldsymbol{Z}_G = \boldsymbol{S}\boldsymbol{Z}_L. \tag{19}$$

Finally, a skip connection operation is adopted to pass the information smoothly within the proposed mechanism

$$\widetilde{\boldsymbol{Z}} = \beta\boldsymbol{Z}_G + \boldsymbol{Z}_L \tag{20}$$

where β denotes the scale parameter, which is initially set as 0 and will be learned while the network training. Technically, self-correlated learning smoothly considers the node correlation in the graph at both the local and global perspective level and passes the graph feature smoothly, which has a potential benefit on refining the consensus latent representations,reconstructing the graph (introduced in Section III-D), and subsequent node clustering.

E. Graph Convolutional Embedding Decoder

In our proposed method, we develop two decoders, i.e., graph reconstruction decoder and graph convolutional embedding clustering decoder. Graph reconstruction decoder is used to effectively learn the hidden representation $\boldsymbol{Z}$. We adopt a graph convolutional embedding clustering decoder to cluster the graph node and optimize the hidden representation. It should be noted that the two components benefit from each other.

1) Graph Reconstruction Decoder: The available graph reconstruction decoders only aim to reconstruct the adjacency information, while the spectral information is ignored. Different from most graph reconstruction decoders, the proposed decoder considers not only the adjacency information but also spectral information. We introduce spectral information into the graph reconstruction decoder. Thus, both the adjacency matrix and the weighted attribute matrix are required to reconstruct our network. We can express the reconstructed adjacency matrix by adopting a simple inner product decoder as

$$\widehat{\boldsymbol{A}} = \text{sigmoid}\left(\boldsymbol{Z}_L^T\boldsymbol{Z}_L\right). \tag{21}$$

The adjacency matrix reconstruction loss and the weighted attribute matrix reconstruction loss can be expressed as

$$L_a = \frac{1}{2N}\|\boldsymbol{A} - \widehat{\boldsymbol{A}}\|_F^2 \tag{22}$$

$$L_w = \frac{1}{2N}\|\boldsymbol{A}\boldsymbol{X} - \boldsymbol{Z}_G\|_F^2 \tag{23}$$

where L_a and L_w are the adjacency matrix reconstruction loss and the weighted attribute matrix reconstruction loss, respectively, and the fusion graph reconstruction loss L_r is formulated as

$$L_r = L_a + \gamma L_w \tag{24}$$

where γ is a predefined hyperparameter used to balance the importance of two graph reconstruction loss, i.e., L_a and L_w, we initialize it as 0 and learn its weight while training the network. By minimizing L_r, the reconstruction loss over the adjacency matrix and the weighted attribute matrix is minimized simultaneously.

2) Graph Convolutional Embedding Self-Training Clustering Decoder: As mentioned above, we can encode the transformed graph into a compact representation by a locality preserving graph convolutional autoencoder and self-correlated learning smoothy module. However, the node clustering is not involved in the graph hidden representation optimization. Simultaneously, the main challenge for unsupervised graph clustering is the nonexistence of labels to train the network. Thus, inspired by DEC [31], a self-training strategy is designed in our network, in which the soft labels are employed to supervise the clustering process. The self-training clustering objective is

$$L_c = \text{KL}(\boldsymbol{P}\parallel\boldsymbol{Q}) = \sum_i\sum_j p_{ij}\log\frac{p_{ij}}{q_{ij}} \tag{25}$$

where $\text{KL}(\cdot\parallel\cdot)$ is the Kullback–Leibler divergence to measure the similarity of the two distributions, p_{ij} is the target distribution, and $\boldsymbol{Q}$ is the soft labels distribution that can be used to measure the similarity between cluster center μ_j and node embedding $\widetilde{z}_i$ of the node i. In our network, a student's

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t-distribution is proposed as a kernel to measure the soft clustering assignment distribution, i.e.

$$q_{ij} = \frac{\left(1 + |\widetilde{z}_i - u_j|^2/\upsilon\right)^{-\frac{\upsilon+1}{2}}}{\sum_j'\left(1 + |\widetilde{z}_i - u_{j'}|^2/\upsilon\right)^{-\frac{\upsilon+1}{2}}} \tag{26}$$

where υ is the freedom degree of the Student's t-distribution. With (26), we can obtain the assigning probability of node i to center j. Furthermore, the all-nodes distribution is reflected by the soft assignment matrix $\boldsymbol{Q} \in \mathbb{R}^{N\times C}$.

On the other hand, to increase the cluster assignment confidence, the target distribution p_{ij} is introduced, with which all nodes are derived to get closer to cluster centers. The target distribution p_{ij} is expressed as

$$p_{ij} = \frac{q_{ij}^2/\sum_i q_{ij}}{\sum_j'\left(q_{ij'}^2/\sum_i q_{ij'}\right)}. \tag{27}$$

As described above, we can get a denser $\boldsymbol{Q}$ distribution by minimizing L_c. Namely, it makes the distribution of nodes in the same class more similar, and the distance between nodes in different classes is longer.

F. Joint Loss and Optimization

In our S^{2}LGCC, the overall learning objective is composed of two parts, including the graph reconstruction loss and self-training clustering loss, and the network joint loss is calculated as

$$L = + \underbrace{L_r}_{\text{Reconstruction}} + \underbrace{\lambda L_c}_{\text{Clustering}} \tag{28}$$

where λ a predefined hyperparameter is used to balance the importance of graph reconstruction and node clustering.

In our method, we should update three main parameters by adopting gradient descent, i.e., the coefficient weight $W^{(l)}$, cluster center μ_j, and target distribution p_{ij}. We adopt gradient decent to update the network weight. The gradient of L_c relative to μ_j with the fixed p_{ij} can be expressed as

$$\frac{\partial L_c}{\partial \mu_j} = \frac{\upsilon+1}{\upsilon}\sum_{i=1}^{N}\left(1 + |\widetilde{z}_i - u_j|^2/\upsilon\right)^{-1}\left(q_{ij} - p_{ij}\right)\left(\widetilde{z}_i - \mu_j\right). \tag{29}$$

With a learning rate η, μ_j is updated as

$$\mu_j = \mu_j - \eta\frac{\partial L_c}{\partial \mu_j}. \tag{30}$$

In the same way, the gradients of L_c relative to $\widetilde{z}_i$ are

$$\frac{\partial L_c}{\partial \widetilde{z}_i} = -\frac{\upsilon+1}{\upsilon}\sum_{j=1}^{c}\left(1+|\widetilde{z}_i - u_j|^2/\upsilon\right)^{-1}\left(q_{ij} - p_{ij}\right)\left(\widetilde{z}_i - \mu_j\right). \tag{31}$$

Then, the coefficient weight $W^{(l)}$ is updated by

$$W^{(l)} = W^{(l)} - \alpha\left(\frac{\partial L_r}{\partial W^{(l)}} + \gamma\frac{\partial L_c}{\partial W^{(l)}}\right). \tag{32}$$

The target distribution p_{ij} is the soft label that can be treated as "groundtruth" and should be updated in every T iterations using all embedded nodes. We can obtain c initial centroids $\{\mu_j\}_{j=1}^c$ by adopting k-means clustering on the embedded nodes $\widetilde{\boldsymbol{Z}}$. Then, p_{ij} is updated by (26) and (27), and the label of n_i could be estimated by

$$n_i = \arg\max_j q_{ij}. \tag{33}$$

The network training process will be stopped as the target distribution p_{ij} change between two consecutive updates is less than a fixed δ or the training epochs T reach the upper set limit. The clustering results are presented by the last optimized soft labels distribution $\boldsymbol{Q}$. The detailed learning procedure of S^{2}LGCC is shown in Algorithm 1.

Algorithm 1 Unsupervised Self-Correlated Learning Smoothy Enhanced Locality Preserving Clustering Network (S^{2}LGCC)

Input: Graph $(\mathcal{V}, \mathcal{E}, \boldsymbol{A})$; Cluster number c; Iterations Number T; Target distribution update interval I, Coefficient weight γ, λ; Distribution threshold δ.
Output: Clustering results Q.

Perform graph autoencoder by Eq. (8) and Eq. (10) to obtain the graph hidden presentation $\boldsymbol{Z}$;
Initialize c initial clusters centroid $\mu_j$with k-means clustering based on $\widetilde{\boldsymbol{Z}}$.
for $t = 1$ to T do
    $\boldsymbol{Z}_L$ is updated by Eq. (13);
    The message passing operation $\boldsymbol{Z}_L$, the normalized self-correlation matrix $\boldsymbol{S}$ and the graph convolution embedding $\widetilde{\boldsymbol{Z}}$ are updated by Eq. (13), Eq. (16), and Eq. (18), respectively;
    The soft labels distribution $\boldsymbol{Q}$ is calculated by Eq. (22);
    if $t\%I == 0$ **then**
    The target distribution $\boldsymbol{P}$ is calculated by Eq. (23);
    The self-training clustering loss L_c is calculated according to Eq. (21);
    if $L_c \le \delta$ **break**;
    Calculate joint embedding loss L and update the weight matrices by Adam gradient descent;
end
Obtain the clustering results (final optimized soft labels distribution $\boldsymbol{Q}$ according to Eq. (29).

G. Computational Complexity Analysis

The computational complexity of our S^{2}LGCC method mainly consists of three parts, including locality preserving graph convolutional embedding encoder, self-correlated learning smoothy mechanism, and graph convolutional embedding decoder. Let N be the number of graph nodes, d_i be the ith layer node spectral dimension, $|\mathcal{E}|$ be the number of nonzero entries of the adjacency matrix $\boldsymbol{A}$, c be the number of clusters, and l be the number of graph embedded autoencoder layers. The computational complexity of multilayer

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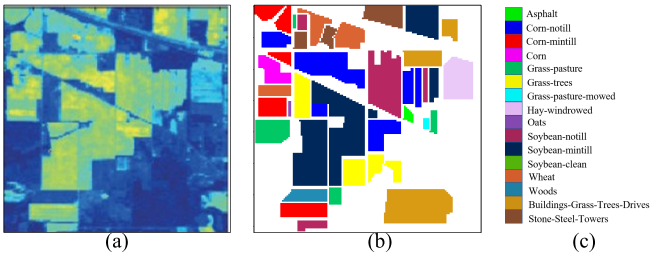

Fig. 3. (a) False color, (b) ground truth, and (c) reference color of IP dataset.

GCN is calculated as $O[(N+|\mathcal{E}|)\sum_{i=1}^{l-1} d_i d_{i-1}]$. For layer-wise GAT, the computational complexity is $O[N(\sum_{i=1}^{l-1} d_i)d_l]$. Therefore, the computational complexity of locality preserving graph convolutional embedding encoder is $O[(N+|\mathcal{E}|)\sum_{i=1}^{l-1} d_i d_{i-1} + N(\sum_{i=1}^{l-1} d_i)d_l]$. For self-correlated learning smoothy mechanism, the computational complexity is $O[(Nd_l)^2 + N(d_l)^2]$. For (28), the main computational complexity is $O[Nc + N\log N]$. Therefore, the main computational complexity of our S^{2}LGCC in each iteration is $O[(N+|\mathcal{E}|)\sum_{i=1}^{l-1} d_i d_{i-1} + N(\sum_{i=1}^{l-1} d_i)d_l + (Nd_l)^2 + N(d_l)^2 + (Nc + N\log N)]$. If our model iterates T times, the overall computational complexity of S^{2}LGCC is $O[\,[(N+|E|)\sum_{i=1}^{l-1} d_i d_{i-1} + N(\sum_{i=1}^{l-1} d_i)d_l + (Nd_l)^2 + N(d_l)^2 + (Nc + N\log N)]T]$.

IV. Experiments

To evaluate the effectiveness of the proposed S^{2}LGCC in large-scale HSI clustering tasks, some experimental comparisons and analyses are shown in this part. Specifically, first, the dataset, experimental parameter settings, and evaluation metrics are introduced (Sections IV-A and IV-B); then, the clustering performances of S^{2}LGCC are compared with ten comparison algorithms in quantitative and visual results (Section IV-C); subsequently, we analyze the parameter selection in the proposed network (Section IV-D); furthermore, the clustering visualization of our method during the network training is demonstrated (Section IV-E); finally, some ablation experiments are implemented to verify the rationality of S^{2}LGCC design (Section IV-F).

A. Benchmark Datasets

In comparative experiments, three well-known HSI datasets are adopted, i.e., Indian Pines (IP), Salinas, and University of Houston 2013 (UH2013) [45]. The IP dataset contains 200 spectral channels with an image size of 145 × 145 and 16 landcover classes. The Salinas dataset is composed of 204 spectral channels with a size of 610 × 340 and nine landcover classes. The UH2013 dataset contains 349 × 1905 pixels with 144 spectral channels and is labeled in 15 categories. Their true-color image, ground-truth map, and reference color are demonstrated in Figs. 3–5, respectively. The details of the three datasets used in the experiment are expressed in Table II.

B. Experimental Setup

The proposed S^{2}LGCC is implemented on an NVIDIA GTX 1080Ti GPU and an i9-10900K CPU with Pytorch 1.7.0 and Python 3.8.5 as a back-end.

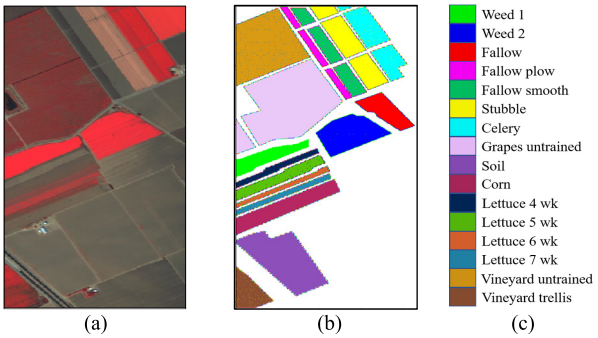

Fig. 4. (a) False color, (b) ground truth, and (c) reference color of Salinas dataset.

To assess the performance of our S^{2}LGCC more intuitively, ten recent state-of-the-art baselines, i.e., k-means [18], FCM [22], FCM-S1 [25], SC [21], FSECSGL [46], S^{5}C [47], SGCNR [48], AE [49] +k-means, DEC [41], and DS^{3}C-Net [37], are considered for comparison. In the investigated methods, k-means, FCM, FCM-S1, and SC are traditional clustering methods designed for natural images; SC, FSECSGL, and S^{5}C are SC methods; SGCNR is based on GCN; AE+k-means, DEC, and DS^{3}C-Net are deep clustering methods. For a fair comparison, the hyperparameters of all experiments are performed the same as the original papers using the same environment.

To qualitatively and quantitatively estimate the clustering performance of S^{2}LGCC, five metrics, including overall accuracy (OA), per-class accuracy (PA), kappa coefficient (κ), normalized mutual information (NMI), and adjusted rand index (ARI), are reported in our experiments. We have independently repeated all experiments 10 times and have taken the average values as the final results.

1) Overall Accuracy: It refers to the percentage of correctly classified samples

$$\mathrm{OA} = \frac{n_{\mathrm{correct}}}{n_{\mathrm{all}}} \tag{34}$$

where n_{correct} is the number of samples with correct clustering, and n_{all} is the number of all samples in clustering.

2) Per-Class Accuracy: It denotes the ratio of the correctly classified pixel in each type

$$\mathrm{PA} = \frac{n^i_{\mathrm{correct}}}{n^i_{\mathrm{all}}} \tag{35}$$

where n^i_{correct} is the number of samples correctly clustered in class i, and n^i_{all} is the number of all samples clustered in class i.

3) Kappa Coefficient (κ): It denotes a robustness measurement with the degree of agreement.

4) Normalized Mutual Information: It is often utilized to measure the similarity of the two-clustering results A and B, namely

$$\mathrm{NMI}(A,B) = \frac{I(A,B)}{\sqrt{H(A)H(B)}} \tag{36}$$

where $H(A)$ and $H(A)$ are the information entropies of A and B, $I(A,B) = H(A) - H(A|B) = H(B) - H(B|A)$

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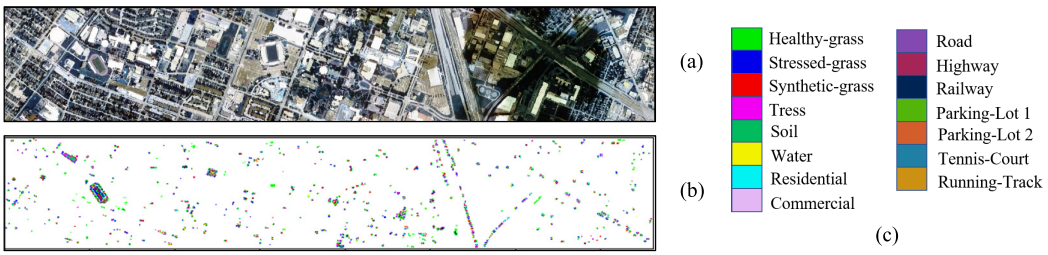

Fig. 5. (a) False color, (b) ground truth, and (c) reference color of UH2013 dataset.

TABLE II
DETAILS OF THE DATASETS USED FOR THE EVALUATION OF THE INVESTIGATED ALGORITHMS

NO.	Indian Pines		Salinas		UH2013	
	Landcover name	Pixels	Landcover name	Pixels	Landcover name	Pixels
1	Asphalt	46	Weed 1	2009	Healthy-grass	1251
2	Corn-notill	1428	Weed 2	3726	Stressed-grass	1254
3	Corn-mintill	830	Fallow	1976	Synthetic-grass	697
4	Corn	237	Fallow-plow	1394	Tress	1244
5	Grass-pasture	483	Fallow-smooth	2678	Soil	1242
6	Grass-trees	730	Stubble	3959	Water	325
7	Grass-pasture-mowed	28	Celery	3579	Residential	1268
8	Hay-windrowed	478	Grapes untrained	11271	Commercial	1244
9	Oats	20	Soil	6203	Road	1252
10	Soybean-notill	972	Corn	3278	Highway	1227
11	Soybean-mintill	2455	Lettuce-4wk	1068	Railway	1235
12	Soybean-clean	593	Lettuce-5wk	1927	Parking-Lot 1	1233
13	Wheat	205	Lettuce-6wk	916	Parking-Lot 2	469
14	Woods	1265	Lettuce-7wk	1070	Tennis-Court	428
15	Buildings-Grass-Trees-Drives	386	Vineyard-untrained	7268	Running-Track	660
16	Stone-Steel-Towers	93	Vineyard-trellis	1807		

is the mutual information of A and B. Obviously, NMI $= 0$, if A and B are independent of each other; NMI $= 1$, if A is the same as B.

5) Adjusted Rand Index: It indicates the similarity between clustering results and ground-truth and can be defined as

$$\mathrm{ARI} = \frac{\mathrm{RI} - E[\mathrm{RI}]}{\max(\mathrm{RI}) - E[\mathrm{RI}]}$$
$$\mathrm{RI} = \frac{a+b}{C_2^{n_{\mathrm{samples}}}} \tag{37}$$

where RI is the raw rand index, and a and b are the logarithm of samples in the same and different category in clustering results and ground-truth, respectively. $C_2^{n_{\mathrm{samples}}}$ is all possible combinations of samples. Obviously, ARI$\in [-1, 1]$.

For our proposed method, seven main parameters, namely, number of superpixels K (graph node), number of graph convolutional embedding encoder layers l, learning rate η, number of epochs T, hyperparameter λ, scale parameter β, and distribution threshold δ, should be preset. We show their optimal values in Table III and will analyze their sensitivity testing in Section IV-D. The proposed method optimization stops as the target distribution p_{ij} change between two consecutive updates is less than a fixed δ or the training epochs T reach the upper set limit. It should be noted that the setting of δ is very small and will not be triggered under normal circumstances, except for the discrete case of clustering.

TABLE III
OPTIMAL HYPERPARAMETERS PRESET IN S2LGCC

Dataset	l	K	T	η	λ	β	δ
IP	4	1000	200	0.001	10	3	0.1%
Salinas	4	7000	200	0.001	10	3	0.1%
UH 2013	4	14000	200	0.001	10	3	0.1%

C. Quantitative and Qualitative Performances Comparison

In this section, we perform clustering on the IP, Salinas, and UH2013 datasets and report the quantitative clustering results in detail in Tables IV–VI, where the best performances and the second-best performances are marked in bold and underlined, respectively. Furthermore, the corresponding qualitatively visual clustering results are shown in Figs. 6–8.

1) Clustering Results on IP Dataset: We perform the first experiment on the IP dataset to evaluate the clustering performance of the proposed S^{2}LGCC for a small-scale HSI dataset with severe noise interference. The quantitative clustering accuracy results and the corresponding qualitatively visual clustering maps of 11 algorithms are demonstrated in Table IV and Fig. 6, respectively.

From Table IV, due to extreme imbalance of clustering samples, lack of training labels, and severe noise interference, the comparison methods achieve poorly on IP image, but our S^{2}LGCC still achieves 71.76% in terms of OA, while κ,

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TABLE IV

QUANTITATIVE EXPERIMENTAL PERFORMANCES OF INVESTIGATED METHODS ON IP. BOLD DENOTES THE BEST PERFORMANCES

No.	k-means	FCM	FCM-S1	SC	FSECSGL	S^5C	SGCNR	AE+ k-means	DEC	DS^3C-Net	S^2LGCC
1	0.0594	0.0567	0.0634	0.0000	0.0625	0.1125	0.0000	0.0870	0.0000	0.1142	**0.1962**
2	0.4563	0.5226	0.4426	0.0273	0.2139	0.2537	0.4398	0.1408	0.1877	0.2709	**0.6787**
3	0.1819	0.2118	0.1781	0.0048	0.3216	0.3576	0.0133	0.1940	0.2892	**0.6985**	0.6139
4	0.1237	0.1058	0.2109	0.0000	0.3018	0.1531	0.1477	0.1730	0.2405	0.8861	**0.8926**
5	0.3447	0.2946	0.3393	0.0166	0.5749	0.6279	0.4928	0.5217	0.0000	**0.6534**	0.4510
6	0.6901	0.7645	0.8008	0.0000	0.5180	0.5524	0.2123	0.9329	0.9726	0.9795	**1.0000**
7	0.0000	0.1043	0.0000	0.0000	0.4297	**0.7019**	0.0000	0.0000	0.0000	0.0000	0.1301
8	0.6239	0.8922	0.5706	**1.0000**	0.6139	0.5124	0.9582	0.7134	0.6464	0.9137	**1.0000**
9	0.0000	0.6400	0.0215	0.0000	0.0000	0.1329	0.1500	0.0000	0.0000	0.0000	**0.6775**
10	0.5654	0.2528	0.5268	0.0031	0.3162	0.2618	0.3735	0.5021	0.3693	0.5327	**0.6881**
11	0.4733	0.5141	0.5966	**0.9609**	0.4802	0.2235	0.6933	0.3752	0.3202	0.2957	0.4376
12	0.2592	0.2788	0.2553	0.0017	0.0971	0.0000	0.1754	0.1551	0.2749	0.6572	**0.6813**
13	0.3276	0.4289	0.2956	0.1463	0.6095	0.7531	0.9610	0.5024	0.3610	**1.0000**	**1.0000**
14	0.8559	0.8006	0.9173	**1.0000**	0.6573	0.4629	0.4877	0.7605	0.7763	0.6854	0.9073
15	0.2007	0.2869	0.2584	0.0000	0.1819	0.0000	0.1580	0.6995	0.7021	**1.0000**	**1.0000**
16	0.0000	0.0632	0.0000	0.7419	0.5732	0.7726	0.6882	0.5591	0.6022	0.1986	**0.7729**
OA	0.3811	0.3757	0.3589	0.4153	0.4236	0.4072	0.4530	0.4458	0.4171	0.5738	**0.7176**
κ	0.3572	0.2975	0.2782	0.2794	0.3517	0.3518	0.3730	0.3847	0.3591	0.5364	**0.6754**
NMI	0.4386	0.4380	0.4217	0.5041	0.4231	0.4317	0.4628	0.4527	0.4787	0.6928	**0.7371**
ARI	0.2195	0.2295	0.2070	0.2455	0.2247	0.2296	0.2571	0.2661	0.2968	0.4350	**0.5165**

TABLE V

QUANTITATIVE EXPERIMENTAL PERFORMANCES OF INVESTIGATED METHODS ON SALINAS. BOLD DENOTES THE BEST PERFORMANCES. "OOM" DENOTES OUT OF MEMORY

No.	k-means	FCM	FCM-S1	SC	FSECSGL	S^5C	SGCNR	AE+ k-means	DEC	DS^3C-Net	S^2LGCC
1	0.9985	0.4179	0.4860	OOM	0.1179	0.0000	0.0000	0.0000	0.9000	0.8761	**1.0000**
2	0.5698	0.5524	0.5637	OOM	0.8305	0.9628	0.6402	**1.0000**	0.7933	0.9473	**1.0000**
3	0.9741	0.8501	**1.0000**	OOM	0.2179	0.0434	0.3333	0.5501	0.0000	**1.0000**	**1.0000**
4	0.9865	0.0000	0.9537	OOM	0.9018	0.8512	**0.9950**	0.9340	0.5079	0.6502	0.7312
5	0.7856	0.5540	0.7365	OOM	0.6290	0.7526	0.6830	0.7367	0.7412	**0.8490**	0.7036
6	0.9952	0.9990	0.9704	OOM	0.9729	0.9832	0.9987	0.8406	0.8366	0.7614	**1.0000**
7	0.4810	0.0000	0.0003	OOM	0.5412	0.1126	**0.9798**	0.3328	0.6943	0.5236	0.8249
8	0.7309	0.7772	0.8412	OOM	0.5133	0.4917	0.5478	0.5008	0.8565	0.4608	**0.8639**
9	0.9750	0.6969	0.9602	OOM	0.6817	0.6826	0.7496	0.9997	0.9994	0.6602	**1.0000**
10	0.6169	0.5507	0.5429	OOM	0.2036	0.3158	**0.8959**	0.5863	0.3350	0.8133	0.8032
11	0.3370	0.0000	0.3013	OOM	0.4892	0.6213	0.0962	0.0000	0.0000	**0.8315**	0.6529
12	0.0273	0.0047	0.0000	OOM	0.6208	0.5184	0.8311	0.9460	0.8293	**1.0000**	0.8824
13	0.3548	0.8134	0.4613	OOM	0.0000	**0.9321**	0.0000	0.0000	0.0000	0.0000	0.7241
14	0.5584	0.6512	0.7469	OOM	0.0000	**0.8725**	0.6649	0.9645	0.0000	0.0000	0.5078
15	0.4994	0.4279	0.5064	OOM	0.5931	0.3948	0.0000	0.6237	0.0579	0.5687	**0.7563**
16	0.0000	0.8321	0.8433	OOM	0.8872	0.6617	0.8630	0.9917	0.9092	**0.9182**	0.8806
OA	0.6799	0.5673	0.6318	OOM	0.5606	0.5168	0.6941	0.6568	0.6257	0.6673	**0.8261**
κ	0.6572	0.5431	0.6202	OOM	0.5447	0.5035	0.7017	0.6206	0.5796	0.6326	**0.8145**
NMI	0.7356	0.6859	0.7089	OOM	0.6629	0.6327	0.7986	0.7720	0.7705	0.7785	**0.8613**
ARI	0.5466	0.4693	0.5119	OOM	0.4706	0.4359	0.5922	0.5548	0.5824	0.5531	**0.6501**

NMI, and ARI are 67.54%, 73.71%, and 51.65%, respectively, which exceed the comparison methods. It can also be observed that the traditional methods, i.e., k-means, FCM, and FCM-S1, achieve relatively poor clustering performance, because they are mainly designed for natural image clustering, they cannot make good use of the feature information of HSIs. Although SC, FSECSGL, and S^5C contain a strong ability to express spectral features of HSIs, the clustering accuracies are relatively low. Specifically, their OA performances are 41.53%, 38.28%, and 40.72%, which has no competitive advantage compared with other comparison methods. Compared with shallow methods (i.e., k-means, FCM, FCM-S1, and SC), the deep clustering algorithms (i.e., AE + k-means, DEC, DS^3C-Net, and S^2LGCC) obtain competitive results, which mainly because deep learning can extract high-level features from HSI for clustering. Specifically, compared with the best results (SC 41.53%) in terms of the OA in shallow methods, the clustering accuracies of deep clustering algorithms are increased by 3.05%, 0.18%, and 13.16%. SGCNR can extract the complex nonlinear similarities among pixels by converting high-dimensional spectral features into a graph-cut problem, with which the spatial-spectral can be used for HSI clustering. However, due to the defects of shallow methods, the clustering performances of SGCNR have no potential to

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TABLE VI

QUANTITATIVE EXPERIMENTAL PERFORMANCES OF INVESTIGATED METHODS ON UH2013. BOLD DENOTES THE BEST PERFORMANCES. "OOM" DENOTES OUT OF MEMORY

No.	k-means	FCM	FCM-S1	SC	FSECSGL	S^{5}C	SGCNR	AE+ k-means	DEC	DS^{3}C-Net	S^{2}LGCC
1	0.7926	0.4932	0.4762	OOM	0.4621	0.4813	0.4672	0.8129	0.8185	**0.9235**	0.8462
2	0.1436	0.2831	0.3017	OOM	0.5738	0.2516	0.3628	0.0973	0.7783	0.8702	**1.0000**
3	0.7638	0.0000	0.0000	OOM	0.0872	0.0000	0.6851	**0.9842**	0.4390	0.0000	0.4037
4	0.7024	0.3834	0.3921	OOM	0.5948	0.4420	0.6973	0.7436	0.0000	0.8135	**0.8731**
5	0.0000	0.6932	0.6273	OOM	0.1172	0.0000	0.0726	0.8285	**0.9815**	0.7761	0.6118
6	0.0000	0.0185	0.0372	OOM	0.2560	0.1266	0.1245	**0.7600**	0.0000	0.2118	0.5239
7	0.2416	0.0789	0.0564	OOM	0.1139	0.0510	0.3677	**0.8028**	0.0000	0.3970	0.7028
8	0.0000	0.0949	0.0629	OOM	0.0000	0.1932	0.1025	0.2291	0.2098	0.6290	**0.6619**
9	0.2671	0.1070	0.1324	OOM	0.2138	0.1224	0.3629	0.0000	**0.4744**	0.4702	**0.6330**
10	0.2935	0.2347	0.2179	OOM	0.2952	0.3125	0.2811	0.2771	0.6544	0.5213	**0.6776**
11	0.2613	**0.5417**	0.4386	OOM	0.4891	0.2718	0.3876	0.2777	0.0089	0.3926	0.5097
12	0.7132	0.8021	0.8211	OOM	0.7132	0.8530	0.6979	0.7875	0.5929	0.6211	**0.8221**
13	0.0000	0.0000	0.000	OOM	0.0000	0.0000	0.0000	0.0000	**0.6482**	0.4425	0.5563
14	**1.0000**	0.0000	0.0000	OOM	0.0000	0.0000	0.8143	0.9907	**1.0000**	0.0000	0.6535
15	0.3818	0.0000	0.0427	OOM	0.0000	0.0000	0.4537	0.5727	**0.6258**	0.5233	0.4612
OA	0.3976	0.3070	0.3182	OOM	0.3648	0.3167	0.4462	0.5180	0.4704	0.5473	**0.6382**
κ	0.3641	0.2456	0.2573	OOM	0.3416	0.3221	0.4315	0.4783	0.4278	0.5126	**0.6330**
NMI	0.5217	0.3623	0.3716	OOM	0.4324	0.4078	0.5521	0.5871	0.5660	0.6094	**0.6718**
ARI	0.2638	0.1488	0.1534	OOM	0.2297	0.1726	0.3029	0.3613	0.3330	0.4103	**0.4840**

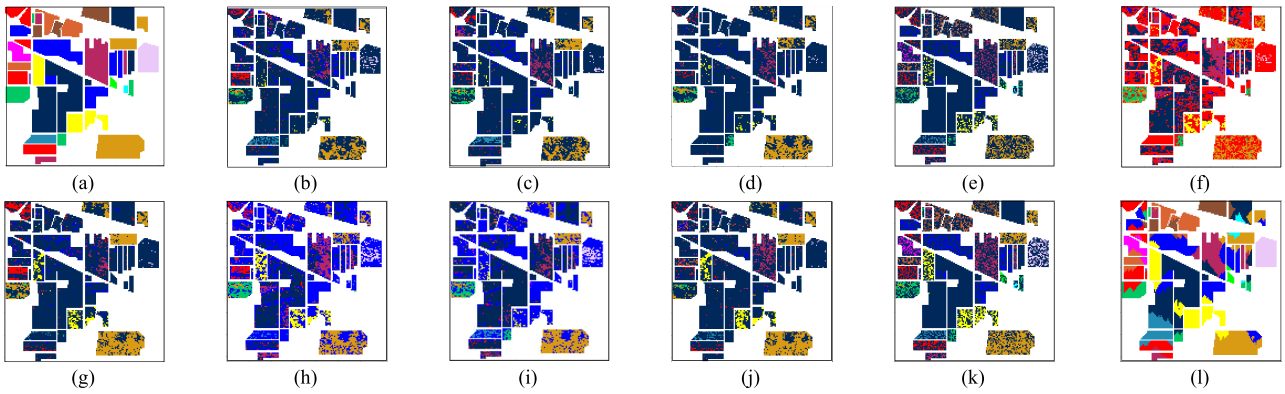

Fig. 6. Clustering maps of investigated methods on IP dataset. (a) Ground-truth map. (b) k-means (OA = 38.11%). (c) FCM (OA = 37.57%). (d) FCM-S1 (OA = 35.89%). (e) SC (OA = 41.53%). (f) FSECSGL (OA = 52.36%). (g) S^{5}C (OA = 40.72%). (h) SGCNR (OA = 45.30%). (i) AE + k-means (OA = 44.58%). (j) DEC (OA = 41.71%). (k) DS^{3}C-Net (OA = 57.38%). (l) S^{2}LGCC (OA = 71.76%).

be further improved. For the qualitatively visual clustering maps in Fig. 6, compared with other competitors, the proposed S^{2}LGCC obtains fewer misclustering and smoother visual effects in the clustering map. For example, the two landcover Hay-windrowed and Soybean–Mintill, k-means and FSECSGL contain lots of "salt and pepper" noises. It should also be observed that the GCN-based method can achieve smoother visual results; however, the clustering map of SGCNR contains many misclassifications. S^{2}LGCC achieves smooth clustering visual results while ensuring a high clustering accuracy.

2) Clustering Results on Salinas Dataset: We perform the second experiment on the Salians dataset to evaluate the performance of our proposed method for middle-scale HSI dataset clustering. The quantitative clustering accuracy results and the corresponding qualitatively visual clustering maps of eleven algorithms are demonstrated in Table V and Fig. 7, respectively.

As the quantitative clustering results are shown in Table V, we can observe that the clustering accuracies of k-means are 67.99%, 65.72%, 73.56%, and 54.66% in terms of OA, κ, NMI, and ARI, respectively, which are the best results of traditional clustering algorithms. It indicates that k-means contains a good clustering effect on a specific HSI dataset with few noises' interferences. It should be noted that "OUT OF MEMORY" occurs in SC for Salinas clustering. This is mainly because SC requires a huge amount of calculation during clustering, resulting in poor adaptability to large datasets. For FSECSGL and S^{5}C, FSECSGL achieves more effective performance, and its OA (56.06%), κ (54.47%), NMI (66.29%), and ARI (47.06%) are improved by 4.38%, 4.12%, 3.02%, and 3.47%, respectively, compared with these values of S^{5}C. Since the GCN-based method can extract the relationship between nodes, the OA, κ, NMI, and ARI achieved by SGCNR are 69.41%, 70.17%, 79.86%, and 59.22%, respectively, which are the second-best results of all investigated algorithms. For the quantitative results obtained by deep clustering methods, different from our consideration, deep clustering methods do not achieve the expected clustering advantages, which illustrates

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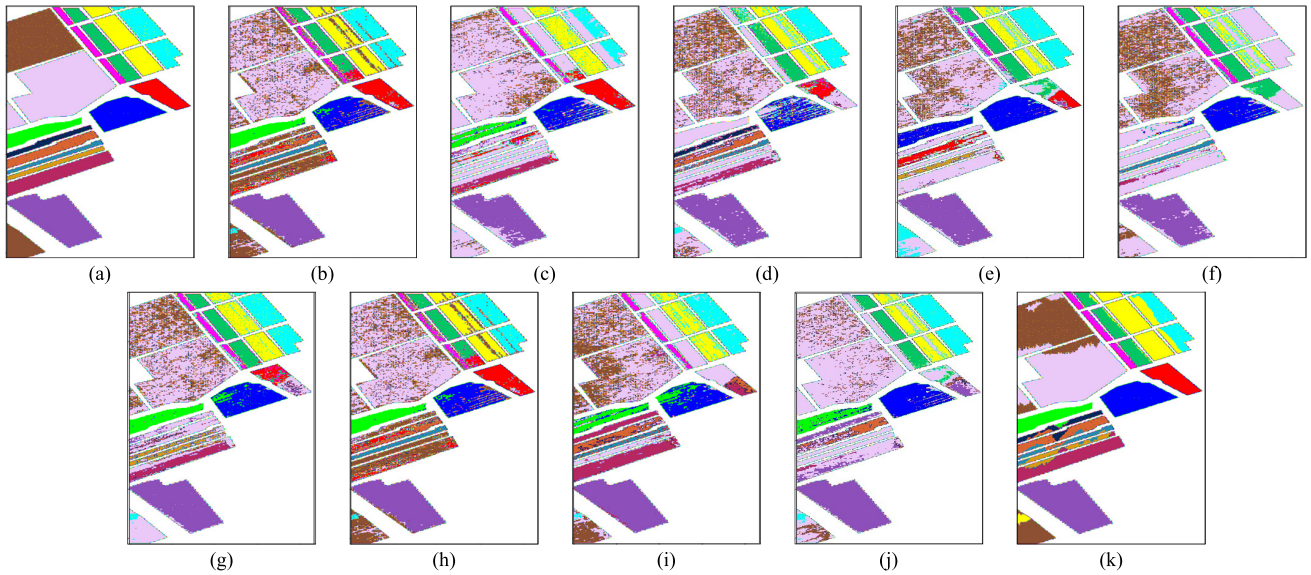

Fig. 7. Clustering maps of investigated methods on Salinas dataset. (a) Ground-truth map. (b) k-means (OA = 67.99%). (c) FCM (OA = 56.73%). (d) FCM-S1 (OA = 63.18%). (e) FSECSGL (OA = 56.06%). (f) S^5C (OA = 51.68%). (g) SGCNR (OA = 69.41%). (h) AE + k-means (OA = 65.68%). (i) DEC (OA = 62.57%). (j) DS^3C-Net (OA = 66.73%). (k) S^2LGCC (OA = 82.61%).

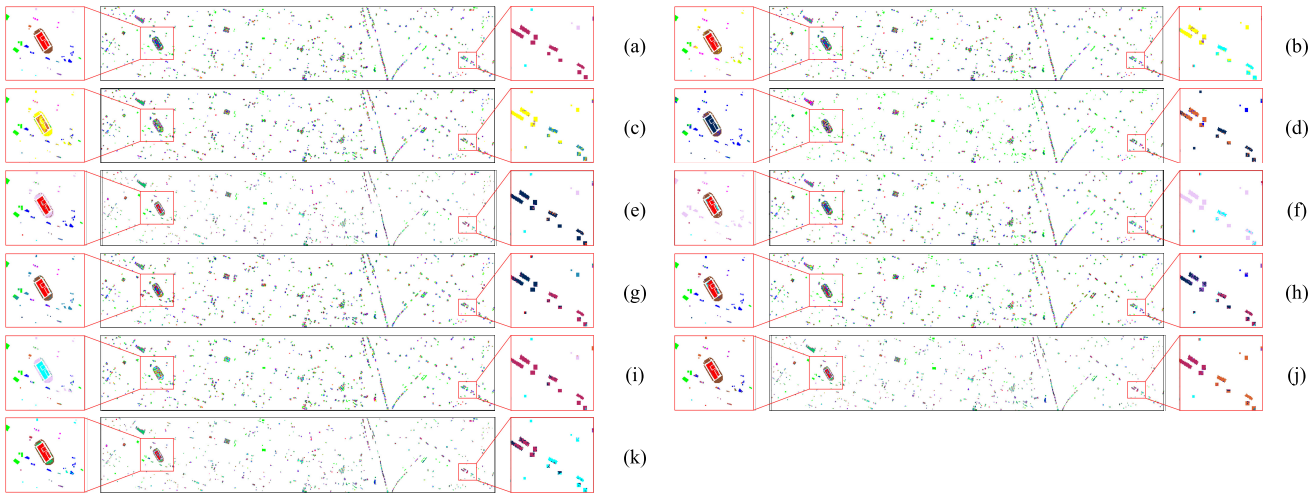

Fig. 8. Clustering maps of investigated methods on UH2013 dataset. (a) Ground-truth map. (b) k-means (OA = 39.76%). (c) FCM (OA = 30.70%). (d) FCM-S1 (OA = 31.82%). (e) FSECSGL (OA = 36.48%). (f) S^5C (OA = 31.67%). (g) SGCNR (OA = 44.62%). (h) AE + k-means (OA = 51.80%). (i) DEC (OA = 47.04%). (j) DS^3C-Net (OA = 54.73%). (k) S^2LGCC (OA = 63.82%).

that deep clustering methods are not always superior to shallow clustering methods. The results of the proposed S^2LGCC are 82.61%, 81.45%, 86.13%, and 65.01% in terms of OA, κ, NMI, and ARI, respectively, which show the advantages of network design. As the qualitatively visual clustering maps of 11 algorithms show in Fig. 7, the misclassification rate is significantly reduced by our S^2LGCC.

3) Clustering Results on UH2013 Dataset: We perform the third experiment on the UH2013 dataset to evaluate the performance of our proposed method for large-scale HSI dataset clustering. The quantitative clustering accuracy results and the corresponding qualitatively visual clustering maps of 11 algorithms are demonstrated in Table V and Fig. 7, respectively.

As the quantitative clustering results are shown in Table V, we can easily summarize that S^2LGCC achieves the best results among all key indicators in the experimental results, which shows that our model has good clustering performances on large-scale HSI datasets. Specifically, the results of the proposed method are improved by 9.09%, 12.04%, 6.24%, and 7.37% in OA, κ, NMI, and ARI, respectively. Compared with traditional methods, i.e., k-means, FCM, and FCM-S1, and SC methods, i.e., FSECSGL and S^5C, do not gain competitive advantages. Specifically, the OAs of the two methods are only 36.48% and 31.67%. Furthermore, SC is suffering from out-of-memory during clustering. It should also be noted that the SGCNR achieves the best results in shallow methods, which verifies the advantages of the GCN mechanism in HSI clustering. Deep learning can extract high-level features for clustering, and the deep clustering algorithms (i.e., AE + k-means, DEC, DS^3C-Net, and S^2LGCC) contain remarkable performances. Specifically, compared with the best results in shallow methods, the clustering accuracies of deep clustering algorithms in terms of the OA are increased by

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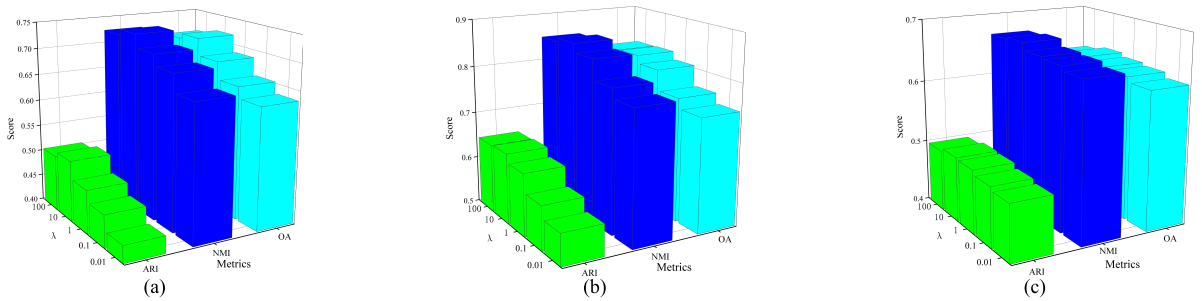

Fig. 9. Sensitivity of S^{2}LGCC with the variation of λ on (a) IP, (b) Salinas, and (c) UH2013.

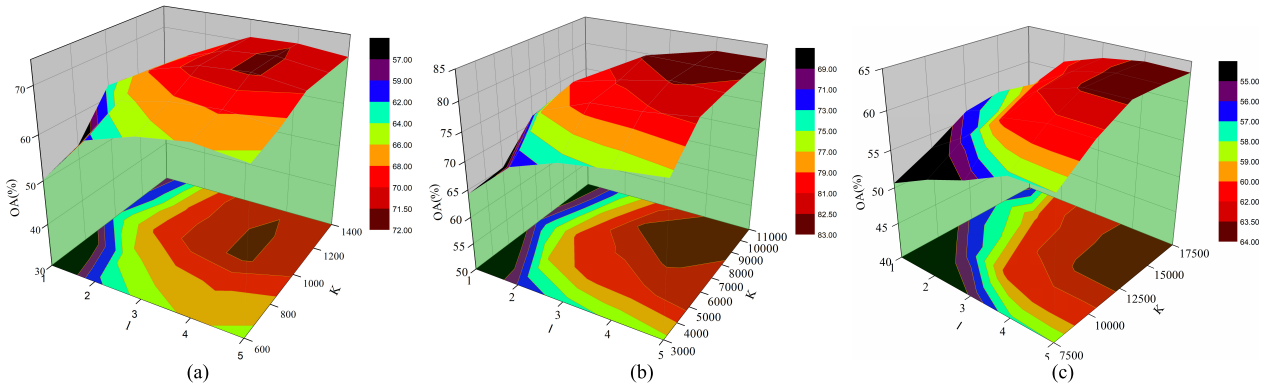

Fig. 10. Sensitivity of S^{2}LGCC with the variation of l and K on (a) IP, (b) Salinas, and (c) UH2013.

7.18%, 2.42%, 10.11%, and 19.2%. As the qualitatively visual clustering maps of 11 algorithms show in Fig. 8, a cleaner classification map can be got by S^{2}LGCC than other comparison methods.

D. Hyperparameter Sensitivity Analysis

As presented in Table III, six hyperparameters (i.e., l, K, T, η, λ, and δ) should be preset for the proposed method. In this section, we investigate the sensitivity of S^{2}LGCC with the variation of λ, l, and K, which have great impacts on clustering results. The experimental setup is consistent with Section IV-B. The grid search strategy is adopted to tune the optimal weights. Furthermore, due to the relationship between hyperparameters, we divide the parameters into two groups, namely, λ shown in Fig. 9, and l and K shown in Fig. 10.

As presented in (28), λ controls the tradeoff between self-training clustering and graph reconstruction. In this part, some experiments are conducted to evaluate the effect of λ on IP, Salinas, and UH2013. The parameter-sensitive results are shown in Fig. 9. In the experiment, λ varies from 0.01 to 100 with other hyperparameters fixed. From the results, we summarize that: 1) λ contains an important influence on clustering results; 2) the trend of main indicators is consistent with the variation of λ; and 3) S^{2}LGCC tends to achieve the best clustering performance when λ is set as 10.

l and K control the depth of the network and the number of graph nodes, respectively. As analyzed in Section III-G, two hyperparameters have important impacts on the computational complexity of S^{2}LGCC. In this section, some experiments are conducted to evaluate the sensitivity of S^{2}LGCC with the variation of l and K on IP, Salinas, and UH2013. The parameter-sensitive results are shown in Fig. 10. In the experiment, K is varied from 600 to 1400 with an interval of 2500 on the IP dataset, from 3000 to 11 000 with an interval of 2000 on the Salinas dataset, and from 7500 to 17 500 on the UH2013 dataset, and L is set from 1 to 5 with other hyperparameters fixed. From the results in Fig. 10, we observe that: 1) more local spatial features of HSI for clustering with a bigger K but more graph node generated; 2) the high-level features of graphs can be extracted with a bigger l. However, more weight parameters and more time are required to train the network; 3) both l and K have a minimum value to achieve the best result; 4) S^{2}LGCC tends to achieve the best clustering performances by setting l to 4 on three datasets; and 5) in terms of K, S^{2}LGCC tends to achieve the best clustering performances with 1000 on IP, 7000 on Salinas, and 14 000 on UH2013.

E. Clustering Visualization

In this section, the distributions of graph nodes in clustering results on three datasets are visualized to intuitively verify the performance of S^{2}LGCC by employing t-SNE [50]. From the maps, the following conclusions can be observed: 1) compared with the original data distributions of graph node, the distance of centroids and the covariance differences between target domains and the source are reduced; 2) misclustering is greatly reduced; and 3) the distance between landcovers from the same class becomes closer, while the distance between landcovers from the different classes is expanded. In a word, the feature

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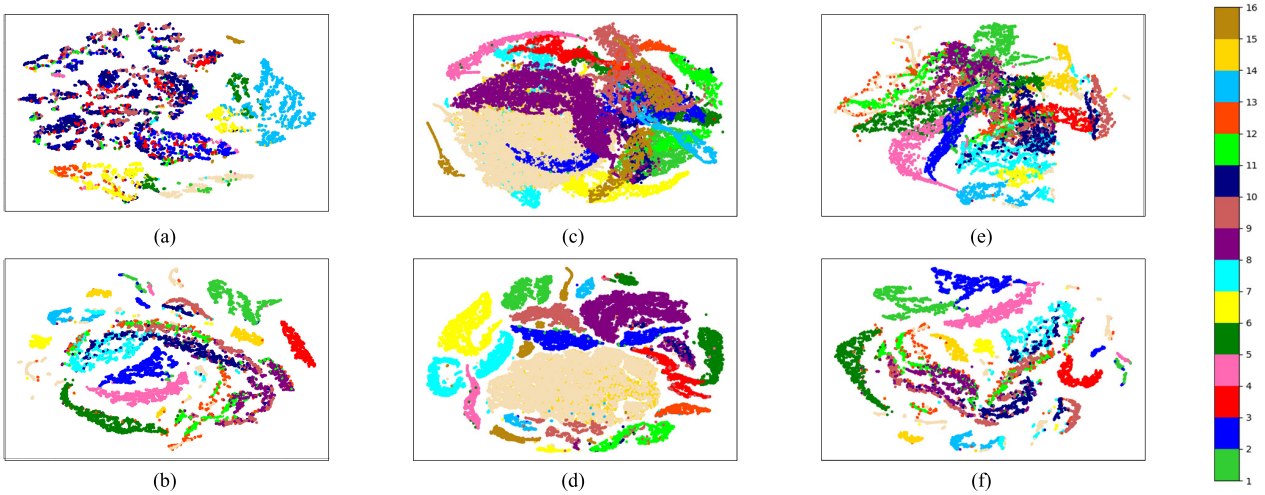

Fig. 11. t-SNE visualization of the S^{2}LGCC on (a) and (b) IP, (c) and (d) Salinas, and (e) and (f) UH2013. (a), (c), and (e) Original distribution of three datasets. (b), (d), and (f) Data distribution after clustering. Different colors in the maps denote different landcovers.

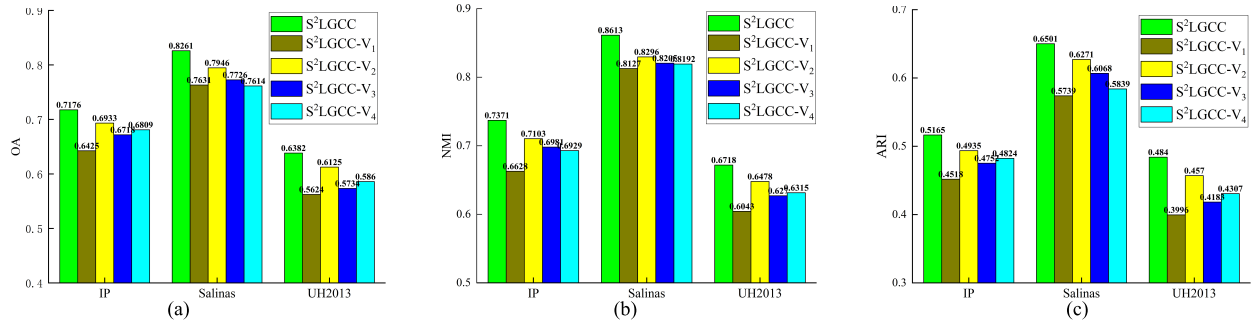

Fig. 12. Ablation study results on IP, Salinas, and UH2013. (a) OA, (b) NMI, and (c) ARI are reported.

representations learned by the proposed method are both intraclass compact and interclass dispense.

F. Ablation Study

To validate the importance of spectral-spatial transformation, LGAT, self-correlated learning smoothy, and self-training clustering decoder to improve the clustering performances of S^{2}LGCC on IP, Salinas, and UH2013 datasets, the ablation experiments are performed. The experimental setup is kept identical to Section IV-B. The OA, NMI, and ARI are employed to report the clustering results of different modules; the results are shown in Fig. 12. For simplicity, we utilize "S^{2}LGCC-V$_1$" to represent the reduced methods without spectral-spatial transformation, and the pixels are treated as the graph nodes. The "S^{2}LGCC-V$_2$" is employed to represent reduced methods without LGAT. The "S^{2}LGCC-V$_3$" is adopted to represent reduced methods without self-correlated learning smoothy, and the inner product decoder is used to reconstruct the graph. The "S^{2}LGCC-V$_4$" is employed to represent the reduced methods without a self-training clustering decoder, and the k-means algorithm is adopted to illustrate the smoothy information $\widetilde{\mathbf{Z}}$. From Fig. 12, we can observe that: 1) compared with the base method, the reduced methods have decreased in all indicators; 2) the decline degrees of indexes in reduced methods are different, which means that each module has a different influence on the clustering results in our method; and 3) each module designed in S^{2}LGCC contributes to clustering performance improving.

V. Conclusion

In this article, we develop a novel deep graph-based clustering framework named S^{2}LGCC to extract representative hidden presentations for unsupervised HSI clustering. Specifically, the spectral-spatial transformation is introduced to transform the original HSI into a graph. The local spectral features and spatial structure are preserved; the number of graph nodes and the computational complexity are reduced. To learn the remarkable and adaptive features for HSI clustering, we develop a locality preserving graph convolutional embedding autoencoder, in which an LGAT mechanism is proposed to preserve adaptive layerwise locality features. To extract the smoothy information and nonlocal relationship contained in the hidden representation space for subsequent graph reconstruction and node clustering, a self-correlated learning smoothy module is designed. We perform the self-training clustering with soft labels generated from the graph embedding. The proposed S^{2}LGCC is jointly optimized by the fusion graph reconstruction loss and self-training clustering loss, and the two benefit each other. The experimental results show that our S^{2}LGCC outperforms the benchmarking approaches and achieves a new state of the art.

Due to prior knowledge deficiency, large spectral variability, and high-dimension of HSI, compared with supervised and semisupervised methods, unsupervised methods have encountered significant difficulties in feature extraction and expression for HSI clustering; thus, the clustering accuracy is still

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poor. However, unsupervised clustering methods should have a bright practical prospect with the development of remote sensing technology. In our future work, we will focus on extracting multiscale features from HSI for clustering. More representative methods will be explored and integrated with GCN, e.g., reinforcement learning, contrastive learning, and meta learning.

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REFERENCES

[1] Y. Ding, X. Zhao, Z. Zhang, W. Cai, N. Yang, and Y. Zhan, "Semi-supervised locality preserving dense graph neural network with ARMA filters and context-aware learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[2] H. Hu, M. Yao, F. He, and F. Zhang, "Graph neural network via edge convolution for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[3] B. Rasti *et al.*, "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[4] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020.

[5] Y. Ding, X. Zhao, Z. Zhang, W. Cai, and N. Yang, "Multiscale graph sample and aggregate network with context-aware learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4561–4572, 2021.

[6] Y. Ding, X. Zhao, Z. Zhang, W. Cai, and N. Yang, "Graph sample and aggregate-attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 9646–9660, Nov. 2021.

[8] D. Yao *et al.*, "Deep hybrid: Multi-graph neural network collaboration for hyperspectral image classification," *Defence Technol.*, early access, Feb. 2022, doi: 10.1016/J.DT.2022.02.007.

[9] W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Asymmetric feature fusion network for hyperspectral and SAR image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 18, 2022, doi: 10.1109/TNNLS.2022.3149394.

[10] Y. Gu, J. Chanussot, X. Jia, and J. A. Benediktsson, "Multiple kernel learning for hyperspectral image classification: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6547–6565, Nov. 2017.

[11] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[12] L. Fang, S. Li, X. Kang, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification via multiscale adaptive sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 12, pp. 7738–7749, Dec. 2014.

[13] Y.-Q. Zhao, L. Zhang, and S. G. Kong, "Band-subset-based clustering and fusion for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 747–756, Feb. 2011.

[14] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 10, 2022, doi: 10.1109/TNNLS.2022.3155114.

[15] Z. Zhang, Y. Cai, and W. Gong, "Evolution-driven randomized graph convolutional networks," *IEEE Trans. Syst., Man, Cybern., Syst.*, early access, Mar. 17, 2022, doi: 10.1109/TSMC.2022.3158276.

[16] Y. Ding *et al.*, "AF2GNN: Graph convolution with adaptive filters and aggregator fusion for hyperspectral image classification," *Inf. Sci.*, vol. 602, pp. 201–219, Jul. 2022.

[17] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.

[18] T. Kanungo, D. M. Mount, N. S. Netanyahu, C. D. Piatko, R. Silverman, and A. Y. Wu, "An efficient *k*-means clustering algorithm: Analysis and implementation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 881–892, Jul. 2002.

[19] C. A. Shah, M. K. Arora, and P. K. Varshney, "Unsupervised classification of hyperspectral data: An ICA mixture model based approach," *Int. J. Remote Sens.*, vol. 25, no. 2, pp. 481–487, Jan. 2004.

[20] Y. Zhong, L. Zhang, and W. Gong, "Unsupervised remote sensing image classification using an artificial immune network," *Int. J. Remote Sens.*, vol. 32, no. 19, pp. 5461–5483, 2011.

[21] A. Ng, M. Jordan, and Y. Weiss, "On spectral clustering: Analysis and an algorithm," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 14, 2001, pp. 1–8.

[22] T.-N. Yang, C.-J. Lee, and S.-J. Yen, "Fuzzy objective functions for robust pattern recognition," in *Proc. IEEE Int. Conf. Fuzzy Syst.*, Aug. 2009, pp. 2057–2062.

[23] Y. Zhong and L. Zhang, "An adaptive artificial immune network for supervised classification of multi-/hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 894–909, Mar. 2012.

[24] A. Paoli, F. Melgani, and E. Pasolli, "Clustering of hyperspectral images based on multiobjective particle swarm optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 12, pp. 4175–4188, Dec. 2009.

[25] S. Chen and D. Zhang, "Robust image segmentation using FCM with spatial constraints based on new kernel-induced distance measure," *IEEE Trans. Syst., Man, Cybern., B, Cybern.*, vol. 34, no. 4, pp. 1907–1916, Aug. 2004.

[26] J. Lin, C. He, Z. J. Wang, and S. Li, "Structure preserving transfer learning for unsupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1656–1660, Oct. 2018.

[27] N. Huang, L. Xiao, J. Liu, and J. Chanussot, "Graph convolutional sparse subspace coclustering with nonnegative orthogonal factorization for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[28] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[29] J. Xu, J. E. Fowler, and L. Xiao, "Hypergraph-regularized low-rank subspace clustering using superpixels for unsupervised spatial–spectral hyperspectral classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 871–875, May 2021.

[30] J. Fan, T. Chen, and S. Lu, "Superpixel guided deep-sparse-representation learning for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 11, pp. 3163–3173, Nov. 2018.

[31] J. Xie, R. Girshick, and A. Farhadi, "Unsupervised deep embedding for clustering analysis," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 478–487.

[32] Y. Li, P. Hu, Z. Liu, D. Peng, J. T. Zhou, and X. Peng, "Contrastive clustering," in *Proc. Conf. Artif. Intell. (AAAI)*, 2021, pp. 8547–8555.

[33] P. Ji, T. Zhang, H. Li, M. Salzmann, and I. Reid, "Deep subspace clustering networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–10.

[34] M. Zeng, Y. Cai, X. Liu, Z. Cai, and X. Li, "Spectral–spatial clustering of hyperspectral image based on Laplacian regularized deep subspace clustering," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2694–2697.

[35] J. Li *et al.*, "A dense connection encoding–decoding convolutional neural network structure for semantic segmentation of thymoma," *Neurocomputing*, vol. 451, pp. 1–11, Sep. 2021.

[36] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[37] J. Lei, X. Li, B. Peng, L. Fang, N. Ling, and Q. Huang, "Deep spatial–spectral subspace clustering for hyperspectral image," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 7, pp. 2686–2697, Jul. 2021.

[38] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[39] Y. Cai *et al.*, "Superpixel contracted neighborhood contrastive subspace clustering network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, doi: 10.1109/TGRS.2022.3179637.

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[40] X. Zhao, L. Liu, Y. Wang, L. Zhang, and A. Plaza, "DS^{4}L: Deep semisupervised shared subspace learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.
[41] J. Bruna, W. Zaremba, A. Szlam, and Y. LeCun, "Spectral networks and locally connected networks on graphs," 2013, *arXiv:1312.6203*.
[42] T. N. Kipf and M. Welling, "Variational graph auto-encoders," 2016, *arXiv:1611.07308*.
[43] B. Schölkopf, A. Smola, and K.-R. Müller, "Nonlinear component analysis as a kernel eigenvalue problem," *Neural Comput.*, vol. 10, no. 5, pp. 1299–1319, Jul. 1998.
[44] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
[45] C. Debes *et al.*, "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[46] X. Yang, G. Lin, Y. Liu, F. Nie, and L. Lin, "Fast spectral embedded clustering based on structured graph learning for large-scale hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[47] S. Matsushima and M. Brbic, "Selective sampling-based scalable sparse subspace clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–10.
[48] R. Wang, F. Nie, Z. Wang, F. He, and X. Li, "Scalable graph-based clustering with nonnegative relaxation for large hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7352–7364, Oct. 2019.
[49] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, 2006.
[50] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 1–27, 2008.

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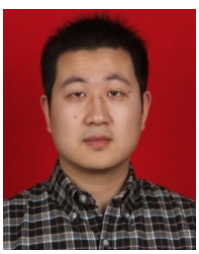

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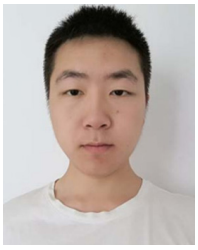

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